Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domains and Ranges of Inverse Trig Functions**

Consider the complex number z such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that a is a real number, find the possible values of a.

Final Answer: ± 1. The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. We are given $z = a + 5i$ and $|z| = \sqrt{26}$. Therefore, $|z| = \sqrt{a^2 + 5^2} = \sqrt{a^2 + 25}$. We are given that $|z| = \sqrt{26}$. So, we have the equation: $\sqrt{a^2 + 25} = \sqrt{26}$. Squaring both sides gives: $a^2 + 25 = 26$. Subtracting 25 from both sides, we get: $a^2 = 1$. Taking the square root of both sides, we find: $a = \pm 1$. Common mistakes include: ignoring the fact that a can be ne.

2 **[Foundational]** • **Target: Derivatives of Inverse Trigonometric Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.333. $P(\text{def}) = 0.360$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.333$

3 **[Foundational]** • **Target: Tangent Lines and Extreme Values**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{3}{10}, \frac{1}{10}, \frac{2}{5}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.

Final Answer: 0.321. $P(\text{def}) = 0.280$ (law of total probability), $P(M_1|\text{def}) = \frac{P(M_1)P(\text{def}|M_1)}{P(\text{def})} \approx 0.321$

4 **[Foundational]** • **Target: Integration Producing Inverse Trig Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{3}{10}, \frac{2}{5}, \frac{1}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.480. $P(\text{def}) = 0.250$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.480$

5 **[Foundational]** • **Target: Domains and Ranges of Inverse Trig Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.

Final Answer: 0.333. $P(\text{def}) = 0.360$ (law of total probability), $P(M_1|\text{def}) = \frac{P(M_1)P(\text{def}|M_1)}{P(\text{def})} \approx 0.333$

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6 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_3|\text{defective})$, to 3 decimal places.

Final Answer: 0.400. $P(\text{def}) = 0.300$ (law of total probability), $P(M_3|\text{def}) = \frac{P(M_3)P(\text{def}|M_3)}{P(\text{def})} \approx 0.400$

7 [Foundational] • Target: Tangent Lines and Extreme Values

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.400. $P(\text{def}) = 0.300$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.400$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration Producing Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{3}{10}, \frac{1}{5}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.194. $P(\text{def}) = 0.310$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.194$

9 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{10}, \frac{1}{5}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.240. $P(\text{def}) = 0.250$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.240$

10 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{10}, \frac{3}{10}, \frac{1}{10}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.

Final Answer: 0.188. $P(\text{def}) = 0.160$ (law of total probability), $P(M_1|\text{def}) = \frac{P(M_1)P(\text{def}|M_1)}{P(\text{def})} \approx 0.188$

11 [Intermediate] • Target: Tangent Lines and Extreme Values

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{2}{5}, \frac{1}{10}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.111. $P(\text{def}) = 0.270$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.111$

12 [Intermediate] • Target: Integration Producing Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{3}{10}, \frac{3}{10}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Final Answer: 0.265. $P(\text{def}) = 0.340$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.265$

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13 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Line 1: $\mathbf{r} = (-2;\ -1;\ -5) + s(-1; 4;\ -1)$. Line 2: $\mathbf{r} = (3;\ -11;\ -12) + t(-2; 3; 4)$. Find their point of intersection.

Final Answer: $(-1,\ -5,\ -4)$. Working through this vectors step:

At $s = -1$ (line 1) and $t = 2$ (line 2), both give the point $(-1,\ -5,\ -4)$.

14 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Line 1: $\mathbf{r} = (-2; 2;\ -1) + s(-1;\ -2; 2)$. Line 2: $\mathbf{r} = (-1; 4;\ -12) + t(0; 0;\ -3)$. Find their point of intersection.

Final Answer: $(-1, 4,\ -3)$. Working through this vectors step:

At $s = -1$ (line 1) and $t = -3$ (line 2), both give the point $(-1, 4,\ -3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 5.01 Further Integration Techniques

Line 1: $\mathbf{r} = (-5;\ -2;\ -2) + s(-2;\ -2; 0)$. Line 2: $\mathbf{r} = (-5;\ -4; 4) + t(1; 2;\ -3)$. Find their point of intersection.

Final Answer: $(-3, 0,\ -2)$. Working through this vectors step:

At $s = -1$ (line 1) and $t = 2$ (line 2), both give the point $(-3, 0,\ -2)$.

16 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 5.02 Differential Equations

Line 1: $\mathbf{r} = (13;\ -13; 1) + s(4;\ -3;\ -1)$. Line 2: $\mathbf{r} = (-11; 8; 1) + t(-4; 4;\ -1)$. Find their point of intersection.

Final Answer: $(1,\ -4, 4)$. Working through this vectors step:

At $s = -3$ (line 1) and $t = -3$ (line 2), both give the point $(1,\ -4, 4)$.

17 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Line 1: $\mathbf{r} = (-1;\ -1; 16) + s(-1; 0; 4)$. Line 2: $\mathbf{r} = (-4;\ -7; 12) + t(-3;\ -3; 4)$. Find their point of intersection.

Final Answer: $(2,\ -1, 4)$. Working through this vectors step:

At $s = -3$ (line 1) and $t = -2$ (line 2), both give the point $(2,\ -1, 4)$.

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18 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

Line 1: $\mathbf{r} = (-3; 2; -3) + s(0; 0; -3)$. Line 2: $\mathbf{r} = (0; -1; 15) + t(-1; 1; -4)$. Find their point of intersection.

Final Answer: $(-3, 2, 3)$. Working through this vectors step:

At $s = -2$ (line 1) and $t = 3$ (line 2), both give the point $(-3, 2, 3)$.

19 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Line 1: $\mathbf{r} = (7; 7; 1) + s(2; 1; -1)$. Line 2: $\mathbf{r} = (0; 8; 7) + t(-1; 4; 3)$. Find their point of intersection.

Final Answer: $(1, 4, 4)$. Working through this vectors step:

At $s = -3$ (line 1) and $t = -1$ (line 2), both give the point $(1, 4, 4)$.

20 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Line 1: $\mathbf{r} = (5; -2; 3) + s(-1; -3; -1)$. Line 2: $\mathbf{r} = (5; -11; 15) + t(0; -3; 4)$. Find their point of intersection.

Final Answer: $(5, -2, 3)$. Working through this vectors step:

At $s = 0$ (line 1) and $t = -3$ (line 2), both give the point $(5, -2, 3)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. **Domain:** $[-1, 1]$**; Range:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$ ✓

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$

To make sine invertible, its domain is restricted to $[-\pi/2, \pi/2]$ with range $[-1, 1]$. Thus $\arcsin x$ has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.

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22 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the domain and range of $y = \arccos x$?

a. **Domain:** $[-1, 1]$**; Range:** $[0, \pi]$ ✓

b. Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

The cosine function is restricted to $[0, \pi]$ where it is strictly decreasing. Therefore $\arccos x$ maps $[-1, 1]$ onto $[0, \pi]$.

23 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the horizontal asymptotes of the graph of $y = \arctan x$?

a. $y = \frac{\pi}{2}$ **and** $y = -\frac{\pi}{2}$ ✓

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

As $x \to \infty$, $\arctan x \to \frac{\pi}{2}$; as $x \to -\infty$, $\arctan x \to -\frac{\pi}{2}$.

24 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1-9x^2}}$ ✓

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{\sqrt{1-3x^2}}$

d. $\frac{3}{1+9x^2}$

By the chain rule: $\frac{d}{dx}[\arcsin(u)] = \frac{u'}{\sqrt{1-u^2}} = \frac{3}{\sqrt{1-(3x)^2}} = \frac{3}{\sqrt{1-9x^2}}$.

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25 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$ ✓

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$

$\frac{d}{dx}[\arctan(x/2)] = \frac{1/2}{1+(x/2)^2} = \frac{1/2}{1+x^2/4} = \frac{1/2}{(4+x^2)/4} = \frac{2}{4+x^2}$.

26 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$ ✓

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

Because $\arcsin x + \arccos x = \frac{\pi}{2}$ for all $x \in [-1, 1]$, differentiating both sides yields $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x]$.

27 [Foundational] • Target: Tangent Lines and Extreme Values

Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1 ✓

b. 0

c. $\pi/2$

d. Undefined

$y'(x) = \frac{1}{\sqrt{1-x^2}}$. At $x = 0$, $y'(0) = \frac{1}{\sqrt{1-0}} = 1$.

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28 [Foundational] • Target: Tangent Lines and Extreme Values

What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?

a. The slope approaches $+\infty$ (vertical tangent) ✓

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

As $x \to 1^-$, $\sqrt{1-x^2} \to 0^+$, so $y'(x) = \frac{1}{\sqrt{1-x^2}} \to +\infty$, resulting in a vertical tangent at the boundary.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent Lines and Extreme Values

Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$ ✓

b. $y - \frac{\pi}{4} = 1(x-1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

Point is $(1, \arctan 1) = (1, \pi/4)$. Derivative is $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$. Tangent line: $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$.

30 [Intermediate] • Target: Integration Producing Inverse Trig Functions

Evaluate $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}dx$.

a. $\frac{\pi}{6}$ ✓

b. $\frac{\pi}{3}$

c. $\frac{\pi}{4}$

d. $\frac{1}{2}$

$[\arcsin x]_0^{1/2} = \arcsin(1/2) - \arcsin(0) = \frac{\pi}{6} - 0 = \frac{\pi}{6}$.

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31 [Intermediate] • Target: Integration Producing Inverse Trig Functions

Evaluate $\int_0^1 \frac{1}{1+x^2}dx$.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. 1

d. $\ln 2$

$[\arctan x]_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$.

32 [Intermediate] • Target: Integration Producing Inverse Trig Functions

Evaluate $\int \frac{1}{\sqrt{4-9x^2}}dx$.

a. $\frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$ ✓

b. $\frac{1}{2}\arcsin\left(\frac{3x}{2}\right) + C$

c. $\arcsin\left(\frac{3x}{2}\right) + C$

d. $\frac{1}{6}\arcsin\left(\frac{3x}{2}\right) + C$

Factor out 4: $\sqrt{4(1-9x^2/4)} = 2\sqrt{1-(3x/2)^2}$. Let $u = 3x/2$, $du = 3/2dx \Rightarrow dx = 2/3du$. Integral: $\frac{1}{2}\cdot\frac{2}{3}\int\frac{du}{\sqrt{1-u^2}} = \frac{1}{3}\arcsin(u) + C = \frac{1}{3}\arcsin(3x/2) + C$.

33 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Line 1: $\mathbf{r} = (0;\ -1;\ -5) + s(4;\ 0;\ -2)$. Line 2: $\mathbf{r} = (2;\ 3;\ -6) + t(-2;\ -4;\ 1)$. Find their point of intersection.

a. $(1, 0, -4)$

b. $(0, -1, -5)$ ✓

c. $(-1, -3, -7)$

d. $(-2, -3, -3)$

Working through this vectors step: At $s = 0$ (line 1) and $t = 1$ (line 2), both give the point $(0, -1, -5)$.

34 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Line 1: $\mathbf{r} = (6;\ -5;\ 1) + s(1;\ -3;\ 1)$. Line 2: $\mathbf{r} = (2;\ 8;\ 2) + t(1;\ -4;\ -4)$. Find their point of intersection.

a. $(3, 4, -2)$ ✓

b. $(2, 3, -3)$

c. $(2, 5, -4)$

d. $(4, 2, 0)$

Working through this vectors step: At $s = -3$ (line 1) and $t = 1$ (line 2), both give the point $(3, 4, -2)$.

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35 [Intermediate] • Target: Tangent Lines and Extreme Values

Line 1: $\mathbf{r} = (6; 4; -1) + s(-2; -1; 1)$. Line 2: $\mathbf{r} = (-1; 0; -2) + t(-1; -1; -4)$. Find their point of intersection.

a. $(2, 2, 1)$

b. $(-2, 0, 0)$

c. $(2, 3, 1)$

d. $(0, 1, 2)$ ✓

Working through this vectors step: At $s = 3$ (line 1) and $t = -1$ (line 2), both give the point $(0, 1, 2)$.

36 [Intermediate] • Target: Integration Producing Inverse Trig Functions

Line 1: $\mathbf{r} = (-3; -2; 5) + s(1; 4; -4)$. Line 2: $\mathbf{r} = (-2; 2; 1) + t(0; 0; 4)$. Find their point of intersection.

a. $(-3, 3, 3)$

b. $(-1, 1, -1)$

c. $(-3, 1, 2)$

d. $(-2, 2, 1)$ ✓

Working through this vectors step: At $s = 1$ (line 1) and $t = 0$ (line 2), both give the point $(-2, 2, 1)$.

37 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Find the value of k for which $(4, 3; 4, k)$ is singular (has no inverse).

a. 1

b. -3

c. **3** ✓

d. 5

Working through this additional-topics step: $\text{Singular} \Leftrightarrow \det = 0 : 4k - 3(4) = 0 \Rightarrow k = \frac{12}{4} = 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

X takes values $0, 1, 2$ with probabilities $\frac{2}{5}, \frac{3}{10}, \frac{3}{10}$. Find $\text{Var}(X)$.

a. $\frac{29}{100}$

b. $\frac{101}{200}$

c. $\frac{69}{100}$ ✓

d. $\frac{59}{200}$

Working through this statistics step: $E(X) = \frac{9}{10},\ E(X^2) = \frac{3}{2} \Rightarrow \text{Var}(X) = \frac{69}{100}$

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39 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A discrete random variable X takes values $2, 3, 4$ with probabilities $\frac{3}{10}, \frac{1}{5}, \frac{1}{2}$ respectively. Find $E(X)$.

a. $\frac{16}{5}$ ✓

b. 6

c. 1

d. 4

Working through this statistics step: $E(X) = \sum xP(X = x) = 2(\frac{3}{10}) + 3(\frac{1}{5}) + 4(\frac{1}{2}) = \frac{16}{5}$

40 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 6.03 Markov Chains (AI HL)

A discrete random variable X takes values $1, 3, 5$ with probabilities $\frac{3}{10}, \frac{3}{10}, \frac{2}{5}$ respectively. Find $E(X)$.

a. $\frac{16}{5}$ ✓

b. 5

c. 0

d. 1

Working through this statistics step: $E(X) = \sum xP(X = x) = (\frac{3}{10}) + 3(\frac{3}{10}) + 5(\frac{2}{5}) = \frac{16}{5}$

41 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 1.01 Proof by Mathematical Induction

A discrete random variable X takes values $3, 4, 5$ with probabilities $\frac{1}{5}, \frac{1}{10}, \frac{7}{10}$ respectively. Find $E(X)$.

a. 7

b. $\frac{9}{2}$ ✓

c. 1

d. 6

Working through this statistics step: $E(X) = \sum xP(X = x) = 3(\frac{1}{5}) + 4(\frac{1}{10}) + 5(\frac{7}{10}) = \frac{9}{2}$

42 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

X takes values $0, 1, 2$ with probabilities $\frac{3}{10}, \frac{1}{5}, \frac{1}{2}$. Find $\mathrm{Var}(X)$.

a. 2

b. $\frac{39}{25}$

c. $\frac{19}{25}$ ✓

d. $\frac{49}{50}$

Working through this statistics step: $E(X) = \frac{6}{5},\ E(X^2) = \frac{11}{5} \Rightarrow \mathrm{Var}(X) = \frac{19}{25}$

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43 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{1}{5}, \frac{3}{5}$. Find $\mathrm{Var}(X)$.

a. $\frac{37}{50}$

b. $\frac{37}{25}$

c. $\frac{167}{100}$

d. $\frac{16}{25}$ ✓

Working through this statistics step: $E(X) = \frac{7}{5},\ E(X^2) = \frac{13}{5} \Rightarrow \mathrm{Var}(X) = \frac{16}{25}$

44 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 1.04 Partial Fractions

A discrete random variable X takes values $3, 4, 5$ with probabilities $\frac{1}{5}, \frac{1}{5}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 7

b. $\frac{22}{5}$ ✓

c. 6

d. 2

Working through this statistics step: $E(X) = \sum xP(X = x) = 3(\frac{1}{5}) + 4(\frac{1}{5}) + 5(\frac{3}{5}) = \frac{22}{5}$

45 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A discrete random variable X takes values $1, 2, 3$ with probabilities $\frac{1}{5}, \frac{1}{10}, \frac{7}{10}$ respectively. Find $E(X)$.

a. $\frac{5}{2}$ ✓

b. 0

c. 3

d. 5

Working through this statistics step: $E(X) = \sum xP(X = x) = (\frac{1}{5}) + 2(\frac{1}{10}) + 3(\frac{7}{10}) = \frac{5}{2}$

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